Faculty of Information Technology, University of Moratuwa
BSc. (Hons) in Information Technology
Bsc.(Hons) in Artificial Intelligence
Computer Network – IN 2511 Level 2, Semester 2 - Lab Sheet 06

Dynamic Routing (RIP) on Sub-netted networks

We have learned how to use dynamic routing (RIP version 01) on previous lab session.

RIP version 01 uses local broadcasts to share routing information and RIPv1 is a classful protocol. Supports only classful routing (Does not support VLSM).

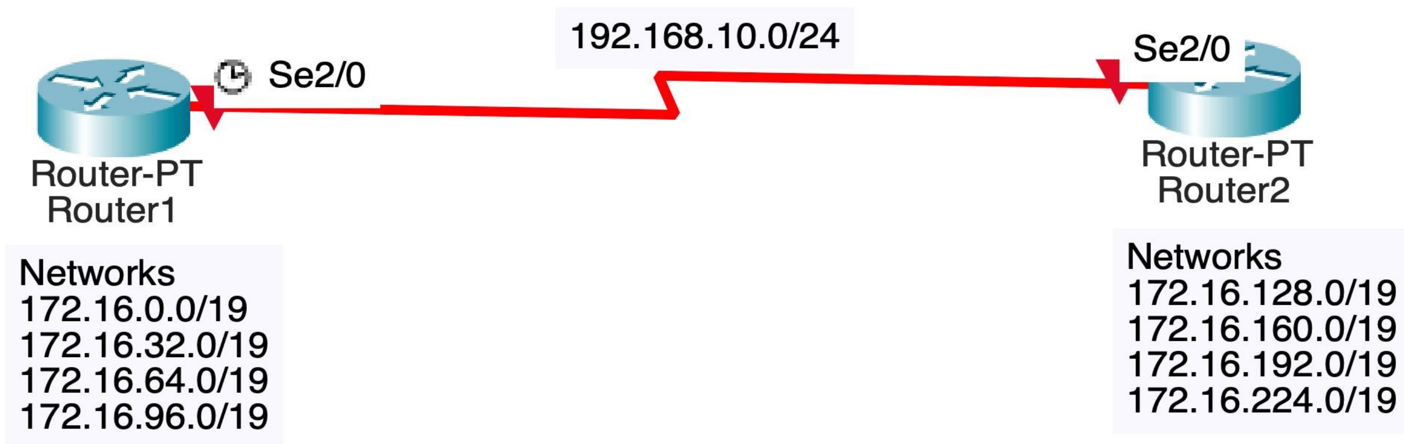
RIP version 02 uses multicasts instead of broadcasts. RIPv2 supports triggered updates. when a change occurs, a RIPv2 router will immediately propagate its routing information to its connected neighbours. RIPv2 is a classless protocol and it supports variable-length subnet masking (VLSM).

Hence, we have to use RIP version 02 on Sub-netted networks.

Although we use RIP version 02 on sub-netted networks, RIP has a feature called auto summarization, which allows Routing Information Protocol (RIP) to summarize its routes to their classful networks automatically.

For an example, consider we are planning to use eight subnets of class B default network 172.16.0.0/16, subnetted using a three-bit subnetting as shown below. If we do a three-bit subnetting on default Class B network 172.16.0.0/16, we will get eight subnets as shown below.

- 172.16.0.0/19
- 172.16.32.0/19
- 172.16.64.0/19
- 172.16.96.0/19
- 172.16.128.0/19
- 172.16.160.0/19
- 172.16.192.0/19
- 172.16.224.0/19



Router1 has four subnets 172.16.0.0/19, 172.16.32.0/19, 172.16.64.0/19, and 172.16.96.0/19 and Router2 has remaining four networks 172.16.128.0/19, 172.16.160.0/19, 172.16.192.0/19 and 172.16.224.0/19 respectivly. After configuring Routing Information Protocol (RIP), view the routing table in Router1, using the show command "show ip route". The output of the show command "show ip route" in Router1 is copied below.

```
Router1>enable
Router1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C      192.168.10.0/24 is directly connected, Serial2/0
       172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
C      172.16.32.0/19 is directly connected, Loopback1
C      172.16.0.0/19 is directly connected, Loopback0
R      172.16.0.0/16 [120/1] via 192.168.10.2, 00:00:05, Serial2/0
C      172.16.96.0/19 is directly connected, Loopback3
C 172.16.64.0/19 is directly connected, Loopback2 Router1#
```

The output of the show command "show ip route" in Router2 is copied below.

```
Router2>enable
Router2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C     192.168.10.0/24 is directly connected, Serial2/0
      172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
C     172.16.160.0/19 is directly connected, Loopback1
C     172.16.128.0/19 is directly connected, Loopback0
C     172.16.224.0/19 is directly connected, Loopback3
C     172.16.192.0/19 is directly connected, Loopback2
R     172.16.0.0/16 [120/1] via 192.168.10.1, 00:00:02, Serial2/0
Router2#
```

Now we can see from the Routing Table of the routers that both the routers are summarizing its subnets to default Class B unsubnetted network 172.16.0.0/16. This will create confusion in network, because both the routers Router1 and Router2 are advertising the same Class B network 172.16.0.0/16. Each router is advertising each other about same network.

To turn off Auto summarization feature in RIP, we can use the "no auto-summary" command from the Router Configuration mode.

Use the following steps in Router1 to disable Auto summarization in Router1 using "no auto-summary" command.

```
Router1>enable
Router1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router1(config)#router rip
Router1(config-router)#version 2
Router1(config-router)#no auto-summary
Router1(config-router)#exit
Router1(config)#exit
Router1#
```

Use the following steps in Router2 to disable Auto summarization in Router2 using "no auto-summary" command.

```
Router2>
Router2>enable
Router2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router2(config)#router rip
Router2(config-router)#version 2
Router2(config-router)#no auto-summary
Router2(config-router)#exit
Router2(config)#exit
Router2#
```

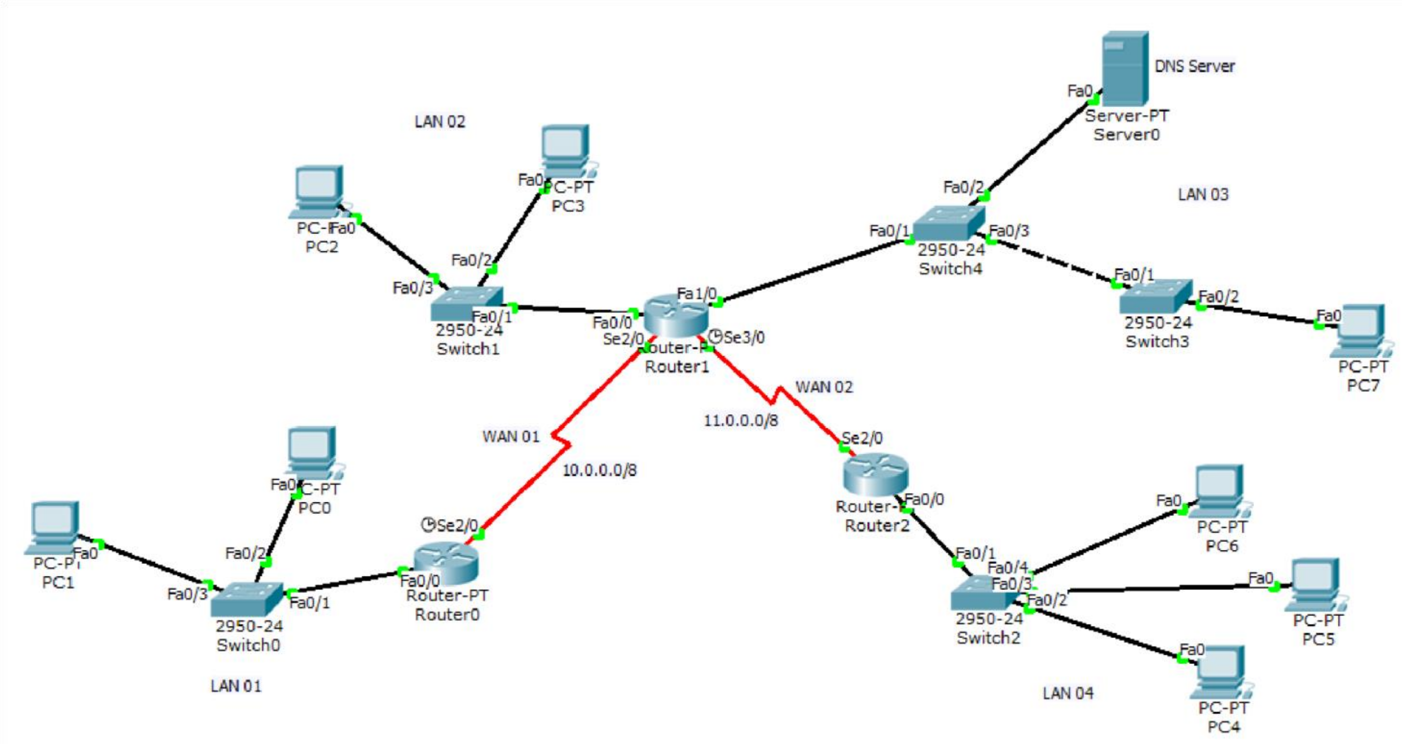
After disabling Auto Summarization using "no auto-summary" command, again check the routing tables in Router1 and Router2 to confirn the routes are advertised individually as separate subnets, using the show command "show ip route" command as shown below.

```
Router1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.10.0/24 is directly connected, Serial2/0
     172.16.0.0/16 is variably subnetted, 9 subnets, 2 masks
R    172.16.128.0/19 [120/1] via 192.168.10.2, 00:00:00, Serial2/0
R    172.16.160.0/19 [120/1] via 192.168.10.2, 00:00:00, Serial2/0
R    172.16.192.0/19 [120/1] via 192.168.10.2, 00:00:00, Serial2/0
R    172.16.224.0/19 [120/1] via 192.168.10.2, 00:00:00, Serial2/0
C    172.16.32.0/19 is directly connected, Loopback1
C    172.16.0.0/19 is directly connected, Loopback0
C    172.16.96.0/19 is directly connected, Loopback3
C    172.16.64.0/19 is directly connected, Loopback2
```

```
Router2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D    - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set
C    192.168.10.0/24 is directly connected, Serial2/0
     172.16.0.0/16 is variably subnetted, 9 subnets, 2 masks
C    172.16.160.0/19 is directly connected, Loopback1
C    172.16.128.0/19 is directly connected, Loopback0
C    172.16.224.0/19 is directly connected, Loopback3
C    172.16.192.0/19 is directly connected, Loopback2
R    172.16.0.0/19 [120/1] via 192.168.10.1, 00:00:05, Serial2/0
R    172.16.32.0/19 [120/1] via 192.168.10.1, 00:00:05, Serial2/0
R    172.16.64.0/19 [120/1] via 192.168.10.1, 00:00:05, Serial2/0
R    172.16.96.0/19 [120/1] via 192.168.10.1, 00:00:05, Serial2/0
```

Exercise 01

Create 4 Local area networks. LAN 01, LAN 02, LAN 03, and LAN 04 as mentioned in the following diagram. You have given a Class B private network address of 172.30.0.0



You need to create 4 subnets for LAN 01, LAN 02, LAN 03, and LAN 04. Find how many minimum number of Host bits taken into the Network bits for this purpose and find the subnet mask. **Use 1st subnet for LAN 1, 2nd subnet for LAN 2, 3rd subnet for LAN 3, and 4th subnet for LAN 4.** Configure the DHCP Servers for address plan of the above LANs according to the following guidelines.

1. In LAN 01 (1st subnet), configure the Router0 as DHCP Server and enable the DHCP on PCs. Take last usable IP Address of 1st subnet as default gateway. Take the IP Address of Server0 as DNS Server.
2. In LAN 2 (2nd subnet), configure the Router1 as DHCP Server and enable the DHCP on PCs. Take last usable IP Address of 2nd subnet as default gateway. Take the IP Address of Server0 as DNS Server.
3. In LAN 3 (3rd subnet), configure the Router1 as DHCP Server and enable the DHCP on PC7 only. Take last usable IP Address of 3rd subnet as default gateway. Take the 1st usable IP Address to the Server0 and Exclude the 1st usable IP address from the DHCP pool. Statically configure the IP Address, subnet mask and default gateway to the Server0.
4. In LAN 4 (4th subnet), configure the Router2 as DHCP Server and enable the DHCP on PCs. Take last usable IP Address of 4th subnet as default gateway. Take the IP Address of Server0 as DNS Server.
5. In WAN 01 Take the 1st usable IP Address to the Se2/0 of Router0 and 2nd usable IP Address to the Se2/0 of Router1. In WAN 02 Take the 1st usable IP Address to the Se3/0 of Router1 and 2nd usable IP Address to the Se2/0 of Router2
6. Configure all the router interfaces and set the clock rate as 64000 for necessary router interfaces.

7. Use **Dynamic routing** in all the routers as per the above scenario.

Exercise 02

Consider the previous exercise 01 with the following requirements.

LAN 01 should support at least 30,000 hosts in the network and other LANs should support at least 8,000 hosts per network. Find the subnet mask for each network and redesign exercise 01 with variable-length subnet masking (VLSM).